

Programming for Economist

PS 3

Class 9

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Plan for today

1. Random numbers and simulations
2. Hints for PS3

1. Random numbers and simulations

- A computer is never truly random but rather pseudo-random
- Reproduce same result through seed/knowledge of stream
 - Reset the seed every time you want the exact same draw/same results.
- Simulation powerful tool to test models and look at results.
- Simulation is the second step after optimisation that we saw last week.

2. Hints for PS3

- Q2
 - Use `plt.hist` and `plt.loglog` to do the plotting
 - Use `"1 - np.arange(1, size + 1) / size"` for the stochastic y when plotting the tail.
- Q3
 - Start with simple logic – use loops!
 - Loop over time and individuals (Here $N=1$, but in Q3 $N>1$)
 - Use if statements to check what state the individual is in, in period t (employed or unemployed)
 - Depending on their draw update their state in $t+1$
 - Jeppe has a more efficient but harder solution – start simple.
- Q4
 - Use your function from Q3
 - Use `np.mean('states', axis=1)` to get mean for each period

2. Hints for PS3

- Q5
 - After filling out the `make_P` function use $\text{pi}_{t+1} = \text{pi}_t @ P$ to do compute next periods `pi`. Make sure `pi_t` and `P` have the right dimensions.
 - Iterate for at least 100 periods and save the `pi` for each period.
 - Make the container for `pi` have dimensions $(T+1, 2)$
- Q6
 - No loops needed, only your knowledge about numpy broadcasting.